

Week 4: Stochastic Optimal Control

The general theory behind the Avellaneda-Stoikov worked example

North Carolina State University | Summer 2026

NC STATE UNIVERSITY

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1. Learning objectives
2. Lecture 1: Dynamic Programming & HJB
3. Lecture 2: Verification & Utilities
4. Illustration
5. Derivation & rationale
6. Getting the data from Coincall
7. Code: week4_merton.py
8. Glossary of key terms
9. Reading
10. References

- ▶ Apply the dynamic programming principle in continuous time
- ▶ Understand the HJB equation and the verification theorem
- ▶ See why CARA / CRRA utilities give tractable problems

- ▶ Controlled diffusions and admissible controls
- ▶ The dynamic programming principle (Bellman)
- ▶ The infinitesimal generator and the HJB equation

- ▶ Verification theorem: a smooth HJB solution IS the value function
- ▶ CARA vs CRRA: where wealth enters the optimal policy
- ▶ Why market making uses CARA

The Stochastic-Control Pipeline



Verification theorem: a smooth HJB solution IS the value function

$$V(t, x) = \sup_u \mathbb{E} \left[\int_t^T f(s, X_s, u_s) ds + g(X_T) \right]$$

Why: The value function is the best achievable expected reward from (t, x) onward. We define the whole problem through V because Bellman's principle then gives a local (differential) equation instead of an intractable global optimization.

$$V(t, x) = \sup_u \mathbb{E}[V(t+dt, X_{t+dt}) + f dt]$$

Why: The dynamic programming principle: an optimal policy is optimal from every later state too. Expanding $V(t + dt, \cdot)$ with Ito's lemma turns this into the HJB PDE.

$$\partial_t V + \sup_u \{ \mathcal{L}^u V + f(t, x, u) \} = 0, \quad V(T, x) = g(x)$$

Why: The HJB equation. $\mathcal{L}^u$ is the generator of the controlled diffusion; the $\sup$ is the instantaneous optimal control. This is the engine behind every model in the course – Avellaneda-Stoikov is one special case.

CARA: $U(w) = -e^{-\gamma w} \Rightarrow$ policy independent of w

Why: Constant absolute risk aversion makes marginal utility scale-free in wealth, so the optimal quotes do not move with the cash balance. CRRA ($U = w^{1-\gamma}/(1-\gamma)$) instead makes the policy scale WITH wealth – natural for investment, awkward for a desk.

- ▶ Theory week – no Coincall data required.
- ▶ Optional: estimate drift/vol from perp klines to parameterize Merton.

Cross-check exact paths at <https://docs.coincall.com/>

```

import torch
# Merton under CARA, solved by backward value iteration on a wealth grid.
w = torch.linspace(-5, 5, 401); dt = 1e-3; mu, r, sig, gam = 0.08, 0.02, 0.2, 1.0
V = -torch.exp(-gam * w) # terminal utility
for _ in range(1000): # march backward in time
    # optimal risky holding  $\pi^* = (\mu - r)/(\text{gam} \cdot \text{sig}^2)$  is wealth-independent (CARA)
    pi = (mu - r) / (gam * sig**2)
    drift = (r*w + pi*(mu - r))
    diff = 0.5 * (pi*sig)**2
    dV = torch.gradient(V, spacing=(w[1]-w[0]).item())[0]
    d2V = torch.gradient(dV, spacing=(w[1]-w[0]).item())[0]
    V = V + dt * (drift*dV + diff*d2V)
print("pi* (constant in wealth):", (mu - r)/(gam*sig**2))

```

Stochastic optimal control choosing controls to optimize an expected objective under randomness

Dynamic programming principle an optimal policy stays optimal from every later state

Value function the best achievable expected reward from a given state onward

Infinitesimal generator operator giving the expected instantaneous change of a function

Verification theorem conditions under which an HJB solution is the value function

CRRA utility constant relative risk aversion; the policy scales with wealth

Admissible control a control satisfying the problem's constraints

- ▶ Pham (2009), Ch. 3-4; Cartea-Jaikumar-Penalva, Ch. 6

- ▶ Pham, H. (2009). Continuous-time Stochastic Control and Optimization with Financial Applications. Springer.
- ▶ Fleming, W. H., & Soner, H. M. (2006). Controlled Markov Processes and Viscosity Solutions (2nd ed.). Springer.
- ▶ Oksendal, B. (2003). Stochastic Differential Equations (6th ed.). Springer.
- ▶ Almgren, R., & Chriss, N. (2001). Optimal execution of portfolio transactions. Journal of Risk, 3(2), 5-40.